

Wish-list

While open-source has a lot of tools to offer the robotics community, there is more to be wished for. Here are some things that the community wishes would be available as open source...

- Custom mechanical hardware
- Sensor technology (especially light detection and ranging or LIDAR sensors)
- Modular robot actuators
- Common input power, common communication and control protocols
- Modular, reconfigurable robot designs

and include several useful code components in C++, as well as software to standardise an unmanned system automatically. Orocos is an open-source tool-chain for real-time robotics that covers all aspects of robot control. Orca is an open-source framework for developing component-based robotic systems. Player is another tool to create software architectures for robotics and sensor systems, and also provides the simulation platforms, Stage and Gazebo, for testing the tools.

The Rossum Project (<http://rossum.sourceforge.net/>) is an attempt to collect, develop and distribute software for robotics applications, and is a great starting point for those looking for open-source tools for their robotic projects.

The journey is just beginning

Summing up the current situation, Jayakrishnan says, "While some specialised tools and platforms are available, currently most of the available open-source robotics components are being borrowed from other fairly mature fields, such as radio-controlled modelling, multimedia, computer gaming, professional toys, etc. Entries to robotic games such as Micromouse and RoboSoccer currently use a lot of open-source components. Another trend is to build a generic platform targeting a particular area, like wheel robots, for instance, and then make the software open-source. Surveyor Corporation's SRV-1 robot is an example."

It appears the open-source trend in robotics is set to grow stronger in the future, and with it, the advancements in robotics. This requires the community to give back as much as it consumes. This can be done by starting new open-source projects and uploading your designs, or by improving the hardware and software belonging to existing projects that you utilise. You can contribute CAD files for hardware components, Gerber files in the case of PCB layouts, the Initial Graphics Exchange Specification (IGES) files for mechanical components, and software code, for new and existing open-source projects.

Azad says, "Right now, the open-source community in robotics is at a very nascent stage, and a lot of organisations are coming up with their own standards,

An open-source model for your business

Being a nascent industry, it would be worthwhile for robotics start-ups to consider an open-source model for their business. This is also quite profitable, as can be seen from the success stories in the software field such as Red Hat and Canonical (Ubuntu).

There are tried and tested business models, specifically for OSH. These include the Forty Percent Model, the Third-Party Catch and Semi-transparent Pricing models. Read more about these at http://www.longtail.com/the_long_tail/2009/01/a-business-mode.html.



T. Jayakrishnan, director and country manager, Energid Robotics and Machine Vision

"Progress in robotics especially, depends on the development of cost-effective, innovative solutions using new-generation OSH platforms and rapid fabrication tools."

so some components and codes are not compatible. In time, there will be one unified platform like Ubuntu or Fedora, which will be universally accepted by all. Once the software is set, the hardware will be developed accordingly, as per the performance of the robot and the types of sensors needed to be integrated. This will help develop a plug-and-play scenario for sensors, actuators, hardware and software."

In all, open source seems to hold a lot of promise for the future. De Sa feels, "The future is bright, but it'll take a while to catch on in India, where people are still wary of releasing their designs to the public. It takes a certain level of mutual trust to say, 'I'll share with you; please share with me'."

At the same time, Jayakrishnan is being rather practical when he concludes that both open source and closed source have roles to play in robotics. "I think it is important to realise that it isn't one over the other, but how both together contribute to the advancement of robotics. Open source is more accessible and lower-cost, but large, centrally organised investments can be made in the closed source space, allowing faster development when there is a great need." 

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